Malware Analysis Playbook (Updated)

Table 1: Malware Analysis Methodology and Key Findings

Phase	Task	Tools	Expected / Observed Key Findings
1. OSINT & Automated Sandboxing	Querying file hashes to gather threat intelligence, threat scores, and automated behavioral reports.	VirusTotal, Hybrid Analysis	Initial detection rates, community insights, network indicators, sandbox behavioral tagging.
2. String & Obfuscation Analysis	Extracting readable and obfuscated strings to identify embedded artifacts.	FLOSS, Strings	Hardcoded QQ email (3905960807@qq.com), registry paths, file artifacts (e.g., Forever_AES_KEY).
3. Basic Static Analysis & Packer Detection	Identifying architecture, compiler information, and packer status.	Detect It Easy, EXEInfo	64-bit Windows executable, not packed (Entropy: 2.90978), compiled using MSVC.
4. Hex & Raw Data Analysis	Verifying magic bytes and inspecting raw binary structure.	HxD	Valid PE structure confirmed (MZ header 4D 5A, DOS stub intact).
5. Advanced Static Analysis (PE Structure)	Examining PE headers, sections, and imported APIs.	PEStudio, CFF Explorer	Cryptographic APIs, filesystem traversal, UI manipulation (SystemParametersInfoW).
6. Dynamic Analysis & Behavioral Monitoring	Executing sample in sandbox and monitoring system, registry, and network behavior.	Process Hacker, Process Monitor, FolderChangesView, RegShot, Wireshark, FakeNet / Fake DNS	Ransomware GUI execution, file encryption activity, registry modifications, network traffic simulation, 100k+ system events.
7. Behavioral Graphing & Visualization	Reconstructing execution flow using collected telemetry.	ProcDot	Process tree mapping, cmd.exe spawning, temp file creation, registry changes.